

Prove that, for the first k odd natural numbers, the sum of these numbers is k^2 .

Base case: $k = 1$

$$1 = 1^2$$

Inductive Hypothesis: Suppose $1 + 3 + \dots + (2k - 1) = k^2$

Inductive step: Show that $\underbrace{1 + 3 + \dots + (2k - 1)}_{k^2} + (2(k+1) + 1) = (k+1)^2$

$$k^2 + 2k + 1 = (k+1)^2$$

$$(k+1)^2 = (k+1)^2$$

Q.E.D.

Critical Note: We adjust $2k+1 \rightarrow 2k-1$ to accommodate the fact that k begins at 1 rather than 0.

Prove that, for the first n natural numbers, their sum is $\frac{n(n+1)}{2}$.

Base case: $n=1$

$$\frac{1(1+1)}{2} = \frac{2}{2} = 1 \quad \checkmark$$

Inductive Hypothesis: Suppose that $1+2+\dots+k = \frac{k(k+1)}{2}$

Inductive step: Show that $1+2+\dots+k+(k+1) = \frac{(k+1)((k+1)+1)}{2}$
 $= \frac{(k+1)(k+2)}{2}$

$$\text{We have } \underbrace{1+2+\dots+k}_{\frac{k(k+1)}{2}} + (k+1) = \frac{k(k+1)}{2} + k+1$$

$$= \frac{k(k+1) + 2(k+1)}{2} = \frac{k^2 + k + 2k + 2}{2} = \frac{k^2 + 3k + 2}{2}$$

$$= \frac{(k+1)(k+2)}{2}$$

$$\therefore 1+2+\dots+n = \frac{n(n+1)}{2}$$

Q.E.D.

$$1+2+3+4+5+6+7+8+9+10$$

$$(1+10) + (2+9) + (3+8) + (4+7) + (5+6)$$

$$\parallel \quad \parallel \quad \parallel \quad \parallel \quad \parallel$$

$\frac{n}{2}$ groups of $n+1$

$$= (n+1) \frac{n}{2} = \frac{n(n+1)}{2}$$